

HMIRM — Human Metacognitive Integrity & Regulation Module

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Independent Methodological Authority

OriginID: OOF-OID-AI-HMIRM-2026-06-03-0001

Architecture Ecosystem: Cognitive Governance Intelligence Architecture (CLIA®)

Architecture Family: Human Cognition Governance

Operational Layer: Human Cognition Integrity Layer

Governed Space: Human Metacognitive Integrity & Regulation

Category: AI & Interpretation

Subcategory: Metacognitive Governance

Type: Human Cognition Integrity Module

Parent Standard: Human Cognition Integrity Standard (HCIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Human Cognition Integrity Standard (HCIS) ·

Cognitive Integrity Standard (CIS) ·

Human-AI Cognition Integrity Standard (HAICS) ·

Multi-Layer Truth Validation Framework (MTVF) ·

Ethical Virtual Integrity Protocol (EVIP) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Human Metacognitive Integrity & Regulation Module (HMIRM) defines the structural conditions under which humans remain capable of monitoring, evaluating, regulating, questioning, and governing their own cognition while interacting with AI systems, information systems, autonomous agents, and future cognition-enhancing technologies.

HMIRM governs metacognitive integrity.

The module ensures that humans remain aware of how they think, why they believe something, where assumptions originate, and when external cognitive systems begin influencing judgment.

A system satisfies HMIRM only if:

- metacognitive awareness remains active
- assumptions remain visible
- cognition remains self-monitorable
- reasoning remains challengeable
- uncertainty remains recognizable

- external cognitive influence remains detectable

A human cognition system that passively accepts external cognition without self-evaluation does not satisfy HMIRM.

Module Operational Role

HMIRM defines the metacognitive governance layer of HCIS.

Its role is to preserve human cognition integrity by ensuring that humans remain conscious participants in their own cognitive processes.

Module Operational Space

HMIRM governs:

- metacognitive awareness
- self-monitoring
- assumption awareness
- cognitive self-regulation
- uncertainty recognition
- reasoning evaluation
- cognitive reflection
- cognition ownership

The module applies wherever humans interact with systems capable of influencing cognition.

Module Function

The module applies wherever systems must preserve:

- self-aware cognition
- critical evaluation
- assumption visibility
- reasoning reflection
- cognitive autonomy
- judgment ownership

Its function is to ensure that humans remain capable of examining their own cognition rather than merely consuming cognitive outputs.

Minimum Implementation Framework

1. Define the Metacognitive Object

The organization must define which cognition processes require metacognitive governance.

This may include:

- decision-making
 - interpretation
 - strategic planning
 - AI-assisted reasoning
 - educational environments
 - professional analysis
 - risk assessment
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- human-AI collaboration

2. Define Metacognitive Conditions

The system must define the conditions under which metacognitive integrity remains valid.

This includes:

- self-awareness requirements
- assumption visibility requirements
- uncertainty recognition requirements
- reflection requirements
- cognition ownership requirements
- evaluation requirements

3. Define Metacognitive Detection Logic

The system must define how metacognitive degradation is identified.

This may include:

- passive acceptance detection
- assumption invisibility detection
- cognitive dependency indicators
- uncertainty suppression detection
- reflection absence detection
- judgment outsourcing indicators

4. Define Operational Response or Governance Logic

The system must define governance logic for metacognitive degradation conditions.

Governance response may include:

- reflection prompts
- uncertainty disclosure
- assumption exposure
- cognition review
- human intervention
- evaluation requirements
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- assumptions
- reasoning evaluations
- uncertainty acknowledgments
- cognition reviews
- governance interventions
- metacognitive assessments

A human cognition system must not remain metacognitively valid if judgment continues while cognition remains unexamined and externally driven.

Use Case 1 — AI-Assisted Executive Decision Environment

Scenario

Executives increasingly rely on AI-generated recommendations, forecasts, and strategic analysis when making high-impact decisions.

Application

HMIRM governs whether decision-makers continue questioning assumptions, evaluating reasoning, and maintaining awareness of cognitive influence.

Result

The organization gains stronger judgment quality and reduced exposure to uncritical acceptance of AI-generated conclusions.

Use Case 2 — Educational AI Learning Environment

Scenario

Students use advanced AI systems for learning, explanation, research, and problem-solving.

Application

HMIRM governs whether learners continue reflecting on their own thinking rather than outsourcing cognition to AI systems.

Result

The educational environment gains stronger independent thinking and reduced cognitive dependency formation.

Canonical Closing Statement

Human Metacognitive Integrity & Regulation Module (HMIRM) defines the structural conditions under which humans remain capable of monitoring, evaluating, regulating, questioning, and governing their own cognition while interacting with AI systems, information systems, autonomous agents, and future cognition-enhancing technologies.

AI fluency without metacognitive regulation creates cognitive dependency, false confidence, and weakened human judgment.

Human cognition remains valid only when it remains capable of governing itself.

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Canonical Source

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